FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Non-Uniform Bodies

Worked solutions 10 questions · 115 marks

EXAM BOARDS Edexcel FM2 9FM0/4C

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*7 marks*	2
Question 2	*8 marks*	4
Question 3	*8 marks*	6
Question 4	*14 marks*	8
Question 5	*11 marks*	11
Question 6	*11 marks*	13
Question 7	*15 marks*	15
Question 8	*12 marks*	17
Question 9	*14 marks*	20
Question 10	*15 marks*	23

1. A thin rod AB of length $3a$ and mass M is non-uniform. At a distance x from A, its mass per unit length is

$$\rho = k\left(a^2 + x^2\right)$$

where k is a positive constant.

(a) Show that $k = \frac{M}{12a^3}$. **[3]**

(b) Find, in terms of a, the distance of the centre of mass of the rod from A. **[4]**

Solution

(a) The mass of the rod satisfies

$$M = \int_0^{3a} \rho \, \mathrm{d}x$$

Substituting $\rho = k\left(a^2 + x^2\right)$:

$$\begin{aligned} M &= \int_0^{3a} k\left(a^2 + x^2\right) \mathrm{d}x \\ &= k\left[a^2 x + \frac{x^3}{3}\right]_0^{3a} \\ &= k\left(3a^3 + \frac{27a^3}{3}\right) \\ &= k\left(3a^3 + 9a^3\right) \\ &= 12ka^3 \end{aligned}$$

Therefore

$$k = \frac{M}{12a^3}$$

as required.

(b) Let $\bar{x}$ be the distance of the centre of mass from A. Then

$$M\bar{x} = \int_0^{3a} x\rho \, \mathrm{d}x$$

The moment of the rod about A is

$$\begin{aligned}\int_0^{3a} x\rho\,\mathrm{d}x &= \int_0^{3a} k\left(a^2x + x^3\right)\mathrm{d}x \\ &= k\left[\frac{a^2x^2}{2} + \frac{x^4}{4}\right]_0^{3a} \\ &= k\left(\frac{9a^4}{2} + \frac{81a^4}{4}\right) \\ &= k\left(\frac{18a^4}{4} + \frac{81a^4}{4}\right) \\ &= \frac{99ka^4}{4}\end{aligned}$$

Using part (a), $M = 12ka^3$, so

$$\begin{aligned}\bar{x} &= \frac{99ka^4}{4} \times \frac{1}{12ka^3} \\ &= \frac{99a}{48} \\ &= \frac{33a}{16}\end{aligned}$$

The centre of mass of the rod is a distance $\frac{33a}{16}$ from A.

2. A non-uniform rod AB has length a. At a distance x from A, the mass per unit length of the rod is $k\left(1+e^{-x/a}\right)$, where k is a positive constant.
Show that the distance of the centre of mass of the rod from A is

$$\frac{(3e-4)a}{2(2e-1)}$$ **[8]**

Solution

Let the mass of the rod be M, and let the centre of mass be at distance $\bar{x}$ from A. For a rod with mass per unit length ρ at distance x from A, the mass and centre of mass satisfy

$$M=\int_0^a \rho\,\mathrm{d}x, \qquad M\bar{x}=\int_0^a x\rho\,\mathrm{d}x$$

Here $\rho = k\left(1+e^{-x/a}\right)$.

First find the mass:

$$\begin{aligned} M &= \int_0^a k\left(1+e^{-x/a}\right)\mathrm{d}x \\ &= k\left[x-ae^{-x/a}\right]_0^a \\ &= k\left[\left(a-ae^{-1}\right)-(0-a)\right] \\ &= k\left(2a-\frac{a}{e}\right) \\ &= \frac{ka(2e-1)}{e} \end{aligned}$$

Now find the moment about A:

$$M\bar{x}=\int_0^a kx\left(1+e^{-x/a}\right)\mathrm{d}x = k\int_0^a x\,\mathrm{d}x + k\int_0^a xe^{-x/a}\,\mathrm{d}x$$

The first integral is

$$\int_0^a x\,\mathrm{d}x=\left[\frac{x^2}{2}\right]_0^a=\frac{a^2}{2}$$

For the second integral, integration by parts is

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv-\int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Take $u=x$ and $\frac{\mathrm{d}v}{\mathrm{d}x}=e^{-x/a}$, so $\frac{\mathrm{d}u}{\mathrm{d}x}=1$ and $v=-ae^{-x/a}$. Then

$$\begin{aligned} \int_0^a xe^{-x/a}\,\mathrm{d}x &= \left[-axe^{-x/a}\right]_0^a+\int_0^a ae^{-x/a}\,\mathrm{d}x \\ &= -\frac{a^2}{e}+\left[-a^2e^{-x/a}\right]_0^a \\ &= -\frac{a^2}{e}+\left(-\frac{a^2}{e}+a^2\right) \\ &= a^2-\frac{2a^2}{e} \end{aligned}$$

Combining the two integrals:

$$\begin{aligned} M\bar{x} &= k\left(\frac{a^2}{2} + a^2 - \frac{2a^2}{e}\right) \\ &= ka^2\left(\frac{3}{2} - \frac{2}{e}\right) \\ &= \frac{ka^2(3e-4)}{2e} \end{aligned}$$

Dividing by the mass:

$$\begin{aligned} \bar{x} &= \frac{ka^2(3e-4)}{2e} \times \frac{e}{ka(2e-1)} \\ &= \frac{(3e-4)a}{2(2e-1)} \end{aligned}$$

as required.

3. A non-uniform straight rod AB has length a. At a distance x from A, the mass per unit length of the rod is $kx\cos\left(\frac{\pi x}{2a}\right)$, where k is a positive constant and $0 \leq x \leq a$.

(a) Show that the mass of the rod is $\frac{2ka^2(\pi-2)}{\pi^2}$. **[3]**

(b) Find, in an exact form, the distance of the centre of mass of the rod from A. **[5]**

Solution

(a) With ρ the mass per unit length, the mass of the rod satisfies

$$M = \int_0^a \rho \, \mathrm{d}x$$

Here $\rho = kx\cos\left(\frac{\pi x}{2a}\right)$, so

$$M = \int_0^a kx\cos\left(\frac{\pi x}{2a}\right)\mathrm{d}x$$

Integration by parts is

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Take $u = x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = \cos\left(\frac{\pi x}{2a}\right)$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ and $v = \frac{2a}{\pi}\sin\left(\frac{\pi x}{2a}\right)$. Then

$$\begin{aligned}\int x\cos\left(\frac{\pi x}{2a}\right)\mathrm{d}x &= \frac{2ax}{\pi}\sin\left(\frac{\pi x}{2a}\right) - \int \frac{2a}{\pi}\sin\left(\frac{\pi x}{2a}\right)\mathrm{d}x \\ &= \frac{2ax}{\pi}\sin\left(\frac{\pi x}{2a}\right) + \frac{4a^2}{\pi^2}\cos\left(\frac{\pi x}{2a}\right) + c\end{aligned}$$

Using $\sin\frac{\pi}{2} = 1$ and $\cos\frac{\pi}{2} = 0$:

$$\begin{aligned}M &= k\left[\frac{2ax}{\pi}\sin\left(\frac{\pi x}{2a}\right) + \frac{4a^2}{\pi^2}\cos\left(\frac{\pi x}{2a}\right)\right]_0^a \\ &= k\left(\frac{2a^2}{\pi} - \frac{4a^2}{\pi^2}\right) \\ &= \frac{2ka^2(\pi-2)}{\pi^2}\end{aligned}$$

as required.

(b) Let $\bar{x}$ be the distance of the centre of mass from A. Then

$$M\bar{x} = \int_0^a x\rho \, \mathrm{d}x$$

so the moment about A needs

$$\int_0^a kx^2\cos\left(\frac{\pi x}{2a}\right)\mathrm{d}x$$

First take $u = x^2$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = \cos\left(\frac{\pi x}{2a}\right)$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = 2x$ and $v = \frac{2a}{\pi}\sin\left(\frac{\pi x}{2a}\right)$:

$$\int x^2\cos\left(\frac{\pi x}{2a}\right)\mathrm{d}x = \frac{2ax^2}{\pi}\sin\left(\frac{\pi x}{2a}\right) - \frac{4a}{\pi}\int x\sin\left(\frac{\pi x}{2a}\right)\mathrm{d}x$$

For the remaining integral take $u = x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = \sin\left(\frac{\pi x}{2a}\right)$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ and $v = -\frac{2a}{\pi}\cos\left(\frac{\pi x}{2a}\right)$:

$$\begin{aligned}\int x\sin\left(\frac{\pi x}{2a}\right)\mathrm{d}x &= -\frac{2ax}{\pi}\cos\left(\frac{\pi x}{2a}\right) + \int \frac{2a}{\pi}\cos\left(\frac{\pi x}{2a}\right)\mathrm{d}x \\ &= -\frac{2ax}{\pi}\cos\left(\frac{\pi x}{2a}\right) + \frac{4a^2}{\pi^2}\sin\left(\frac{\pi x}{2a}\right) + c\end{aligned}$$

Combining these:

$$\int x^2\cos\left(\frac{\pi x}{2a}\right)\mathrm{d}x = \frac{2ax^2}{\pi}\sin\left(\frac{\pi x}{2a}\right) + \frac{8a^2x}{\pi^2}\cos\left(\frac{\pi x}{2a}\right) - \frac{16a^3}{\pi^3}\sin\left(\frac{\pi x}{2a}\right) + c$$

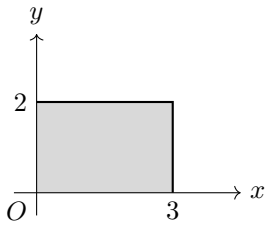
Evaluating from 0 to a, every term is zero at $x = 0$, and at $x = a$ the cosine term is zero:

$$\begin{aligned}M\bar{x} &= k\left(\frac{2a^3}{\pi} - \frac{16a^3}{\pi^3}\right) \\ &= \frac{2ka^3(\pi^2 - 8)}{\pi^3}\end{aligned}$$

Using part (a) for M:

$$\begin{aligned}\bar{x} &= \frac{2ka^3(\pi^2 - 8)}{\pi^3} \times \frac{\pi^2}{2ka^2(\pi - 2)} \\ &= \frac{a(\pi^2 - 8)}{\pi(\pi - 2)}\end{aligned}$$

The centre of mass is $\frac{a(\pi^2-8)}{\pi(\pi-2)} \approx 0.521a$ from A.

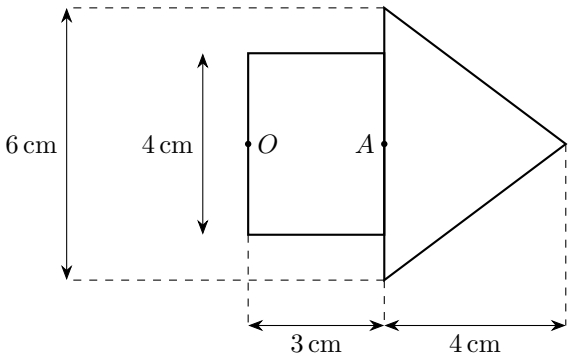


4. The shaded region shown in the diagram above is bounded by the x-axis, the line with equation $x = 3$, the line with equation $y = 2$ and the y-axis.
This region is rotated through 360° about the x-axis to form a solid of revolution. This solid is used to model a **non-uniform** cylinder of length 3 cm and radius 2 cm. The point O is the centre of the end of the cylinder in the plane $x = 0$, and the point A is the centre of the other end of the cylinder.
The mass per unit volume of the cylinder at the point (x, y, z) is $\lambda(12 - x^2)\,\text{kg cm}^{-3}$, where $0 \leq x \leq 3$ and λ is a positive constant.

(a) Show that the mass of the cylinder is $108\pi\lambda$ kg. **[3]**

(b) Show that the centre of mass of the cylinder is 1.25 cm from O. **[4]**

A uniform solid right circular cone has base radius 3 cm, height 4 cm and density $2\lambda\,\text{kg cm}^{-3}$. The cone is attached to the cylinder with the centre of its base in contact with the point A and its axis along the axis of the cylinder, to form the model shown in the diagram below.



The model is placed with the end containing O on a rough plane which is inclined at angle $\alpha°$ to the horizontal. The plane is sufficiently rough to prevent the model from sliding. The model is on the point of toppling.

(c) (i) Show that the centre of mass of the model is 1.75 cm from O. **[4]**

(ii) Find the value of α. **[3]**

Solution

(a) The density $\rho = \lambda(12 - x^2)$ depends only on x. For a solid formed by rotating a region about the x-axis, the mass m_C of the cylinder and the position $\bar{x}_C$ of its centre of mass satisfy

$$m_C = \pi \int_0^3 \rho y^2 \,\mathrm{d}x, \qquad m_C \bar{x}_C = \pi \int_0^3 \rho x y^2 \,\mathrm{d}x$$

Here $y = 2$ throughout, so

$$\begin{aligned} m_C &= \pi \int_0^3 \lambda(12 - x^2) \times 2^2 \, \mathrm{d}x \\ &= 4\pi\lambda \left[12x - \frac{x^3}{3} \right]_0^3 \\ &= 4\pi\lambda(36 - 9) \\ &= 108\pi\lambda \end{aligned}$$

so the mass of the cylinder is $108\pi\lambda$ kg, as required.

(b) By symmetry the centre of mass lies on the x-axis, at $(\bar{x}_C, 0, 0)$. Substituting $y = 2$ and $\rho = \lambda(12 - x^2)$ into the second integral from part (a):

$$\begin{aligned} m_C \bar{x}_C &= 4\pi\lambda \int_0^3 (12x - x^3) \, \mathrm{d}x \\ &= 4\pi\lambda \left[6x^2 - \frac{x^4}{4} \right]_0^3 \\ &= 4\pi\lambda \left(54 - \frac{81}{4} \right) \\ &= 4\pi\lambda \times \frac{135}{4} \\ &= 135\pi\lambda \end{aligned}$$

Using part (a):

$$108\pi\lambda \, \bar{x}_C = 135\pi\lambda \quad \Longrightarrow \quad \bar{x}_C = \frac{135}{108} = 1.25$$

so the centre of mass of the cylinder is 1.25 cm from O, as required.

(c) (i) The volume of the cone is

$$\frac{1}{3}\pi \times 3^2 \times 4 = 12\pi \, \mathrm{cm}^3$$

so its mass is $2\lambda \times 12\pi = 24\pi\lambda$ kg. The centre of mass of a uniform solid cone of height H is $\frac{1}{4}H$ from its base, on its axis, so here it is 1 cm from A, which is $3 + 1 = 4$ cm from O.

By symmetry the centre of mass of the model lies on the axis. Measuring distances from O along the axis:

$$M\bar{x} = \sum m_i x_i, \quad \text{where } M = \sum m_i$$

The total mass is $108\pi\lambda + 24\pi\lambda = 132\pi\lambda$ kg, so

$$\begin{aligned} 132\pi\lambda \, \bar{x} &= 108\pi\lambda \times 1.25 + 24\pi\lambda \times 4 \\ &= 135\pi\lambda + 96\pi\lambda \\ &= 231\pi\lambda \end{aligned}$$

$$\bar{x} = \frac{231}{132} = 1.75$$

so the centre of mass of the model is 1.75 cm from O, as required.

(ii) The axis of the model is perpendicular to the plane. On the point of toppling, the centre of mass G lies vertically above the lowest point P of the rim of the circular end containing O. In the right-angled triangle OGP, $OG = 1.75\,\text{cm}$ along the axis and $OP = 2\,\text{cm}$ along the plane. The line GP is vertical and GO is perpendicular to the plane, so the angle OGP equals the angle of the plane, α°. Hence

$$\tan\alpha^\circ = \frac{OP}{OG} = \frac{2}{1.75} = \frac{8}{7}$$

$$\alpha = 48.8 \qquad (1\text{ d.p.})$$

5. A solid right circular cone has height $2a$ m and base radius a m, and its vertex is V. At a point x m from the plane through V parallel to the base, the density of the cone, ρ kg m^{-3}, is given by

$$\rho = k\left(3 - \frac{x}{a}\right)$$

where k is a positive constant. The mass of the cone is M kg.

(a) Show by integration that $M = \pi k a^3$ and that the centre of mass of the cone is a distance $\frac{7}{5}a$ m from V. **[7]**

(b) A particle of mass m kg is attached to a point A on the circumference of the base of the cone. The cone is freely suspended from V and hangs in equilibrium with its axis making an angle θ with the vertical, where $\tan\theta = \frac{1}{4}$. Find m in terms of M. **[4]**

Solution

(a) Take the axis of the cone as the x-axis, with V at the origin, so that x is the distance from the plane through V parallel to the base. Let y be the radius of the cone at distance x from V. By similar triangles:

$$\frac{y}{x} = \frac{a}{2a} \quad \Longrightarrow \quad y = \frac{x}{2}$$

so the cone is formed by rotating the line $y = \frac{x}{2}$, for $0 \le x \le 2a$, about the x-axis. The density depends only on x. For a solid formed by rotating a region about the x-axis, the mass and centre of mass satisfy

$$M = \pi \int_0^{2a} \rho y^2 \, \mathrm{d}x, \qquad M\bar{x} = \pi \int_0^{2a} \rho x y^2 \, \mathrm{d}x$$

Substituting $y = \frac{x}{2}$ and $\rho = k\left(3 - \frac{x}{a}\right)$:

$$\begin{aligned} M &= \frac{\pi k}{4} \int_0^{2a} \left(3x^2 - \frac{x^3}{a}\right) \mathrm{d}x \\ &= \frac{\pi k}{4} \left[x^3 - \frac{x^4}{4a}\right]_0^{2a} \\ &= \frac{\pi k}{4}\left(8a^3 - 4a^3\right) \\ &= \pi k a^3 \end{aligned}$$

as required.

By symmetry the centre of mass lies on the axis, at distance $\bar{x}$ from V. Substituting into the second integral:

$$\begin{aligned} M\bar{x} &= \frac{\pi k}{4}\int_0^{2a}\left(3x^3 - \frac{x^4}{a}\right)\mathrm{d}x \\ &= \frac{\pi k}{4}\left[\frac{3x^4}{4} - \frac{x^5}{5a}\right]_0^{2a} \\ &= \frac{\pi k}{4}\left(12a^4 - \frac{32a^4}{5}\right) \\ &= \frac{7}{5}\pi k a^4 \end{aligned}$$

Dividing by $M = \pi k a^3$:

$$\bar{x} = \frac{\frac{7}{5}\pi k a^4}{\pi k a^3} = \frac{7}{5}a$$

so the centre of mass is a distance $\frac{7}{5}a$ m from V, as required.

(b) Keep V as the origin with the x-axis along the axis of the cone towards the base, and take the y-axis perpendicular to it, in the plane containing the axis and A. The cone's centre of mass is at $\left(\frac{7}{5}a, 0\right)$, and the particle at A is at $(2a, a)$. Let the centre of mass G of the combined body be at $(\bar{x}_G, \bar{y}_G)$, where

$$\left(\sum m_i\right)\bar{x}_G = \sum m_i x_i, \qquad \left(\sum m_i\right)\bar{y}_G = \sum m_i y_i$$

This gives

$$(M+m)\bar{x}_G = M \times \frac{7}{5}a + m \times 2a$$

$$(M+m)\bar{y}_G = M \times 0 + m \times a = ma$$

When the body is freely suspended from V, G lies vertically below V, so θ is the angle between the axis and VG. Hence

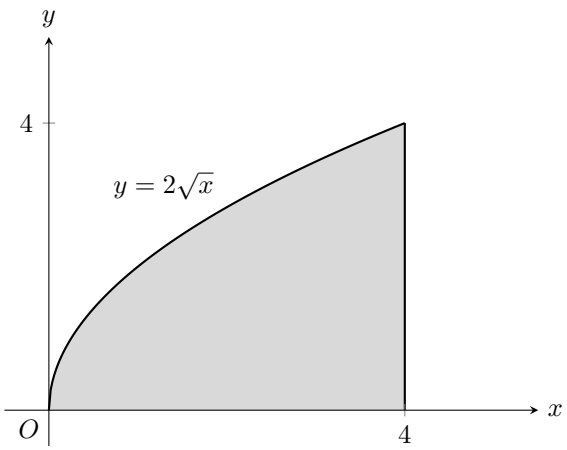
$$\tan\theta = \frac{\bar{y}_G}{\bar{x}_G} = \frac{ma}{\frac{7}{5}Ma + 2ma} = \frac{m}{\frac{7}{5}M + 2m}$$

Setting this equal to $\frac{1}{4}$:

$$4m = \frac{7}{5}M + 2m$$

$$2m = \frac{7}{5}M$$

$$m = \frac{7}{10}M$$



6. The shaded region shown in the diagram above is bounded by the x-axis, the line with equation $x = 4$ and the curve with equation $y = 2\sqrt{x}$, where x and y are measured in centimetres. This region is rotated through 360° about the x-axis to form a solid of revolution, which is used to model a solid paraboloid S with vertex O, height 4 cm and plane circular face of radius 4 cm.
The solid S is non-uniform and its mass per unit volume at the point (x, y, z) is $\lambda(6 - x)\,\text{kg cm}^{-3}$, where $0 \leq x \leq 4$ and λ is a positive constant.

(a) Find the distance of the centre of mass of S from O. **[6]**

A toy is made by joining the plane face of S to the plane face of a uniform solid hemisphere of radius 4 cm and density $\rho\,\text{kg cm}^{-3}$, so that the centres of the two plane faces coincide.
When the toy is placed on a smooth horizontal plane with any point of the curved surface of the hemisphere in contact with the plane, the toy remains at rest.

(b) Find λ in terms of ρ. **[5]**

Solution

(a) By symmetry, the centre of mass of S lies on the x-axis. Let S have mass m_S and centre of mass at $(\bar{x}_S, 0, 0)$. The density depends only on x. For a solid formed by rotating a region about the x-axis, the mass and centre of mass satisfy

$$m_S = \pi \int_0^4 \lambda(6 - x)\, y^2 \,\mathrm{d}x, \qquad m_S \bar{x}_S = \pi \int_0^4 \lambda(6 - x)\, x\, y^2 \,\mathrm{d}x$$

On the curve, $y^2 = 4x$. The mass is

$$\begin{aligned} m_S &= 4\pi\lambda \int_0^4 \left(6x - x^2\right)\mathrm{d}x \\ &= 4\pi\lambda \left[3x^2 - \frac{x^3}{3}\right]_0^4 \\ &= 4\pi\lambda \left(48 - \frac{64}{3}\right) \\ &= \frac{320\pi\lambda}{3}\,\text{kg} \end{aligned}$$

Taking moments about O:

$$\begin{aligned} m_S \bar{x}_S &= 4\pi\lambda \int_0^4 \left(6x^2 - x^3\right) \mathrm{d}x \\ &= 4\pi\lambda \left[2x^3 - \frac{x^4}{4}\right]_0^4 \\ &= 4\pi\lambda(128 - 64) \\ &= 256\pi\lambda \end{aligned}$$

Therefore

$$\bar{x}_S = \frac{256\pi\lambda}{\frac{320\pi\lambda}{3}} = \frac{768}{320} = 2.4$$

The centre of mass of S is 2.4 cm from O.

(b) Let C be the common centre of the two plane faces, the point $(4, 0, 0)$. Whichever point of the hemisphere's curved surface touches the plane, the normal reaction acts vertically along a radius of the hemisphere, so it passes through C. The toy is in equilibrium, so its weight, acting through its centre of mass G, must act along the same vertical line as the reaction. As this holds for every position of the toy, G must be at C, so the centre of mass of the toy is at $x = 4$.

The hemisphere has mass

$$m_H = \frac{2}{3}\pi \times 4^3 \times \rho = \frac{128\pi\rho}{3} \text{ kg}$$

The centre of mass of a uniform solid hemisphere of radius R is $\frac{3}{8}R$ from its plane face, which here is $\frac{3}{8} \times 4 = 1.5$ cm from C, on the side away from O. So the hemisphere's centre of mass is at $x = 5.5$, and that of S is at $x = 2.4$ by part (a).

For the toy, with total mass M and centre of mass at $(\bar{x}, 0, 0)$:

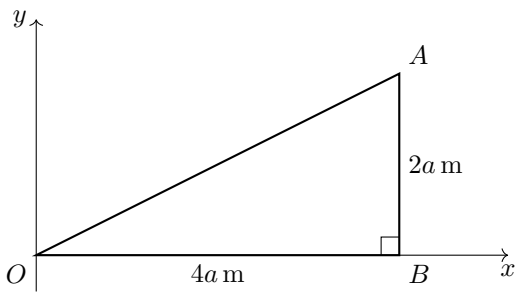
$$M\bar{x} = \sum m_i x_i, \quad \text{where } M = \sum m_i$$

With $\bar{x} = 4$:

$$\left(\frac{320\pi\lambda}{3} + \frac{128\pi\rho}{3}\right) \times 4 = \frac{320\pi\lambda}{3} \times 2.4 + \frac{128\pi\rho}{3} \times 5.5$$

Multiplying through by $\frac{3}{\pi}$:

$$\begin{aligned} 1280\lambda + 512\rho &= 768\lambda + 704\rho \\ 512\lambda &= 192\rho \\ \lambda &= \frac{3\rho}{8} \end{aligned}$$



7. A triangular lamina OAB of mass M kg has a right angle at B, with $OB = 4a$ m and $AB = 2a$ m. The diagram above shows the lamina in the x–y plane with origin O, with OB along the x-axis and BA parallel to the y-axis.
The mass per unit area, ρ kg m^{-2}, of the lamina at a point a distance y m from the x-axis is given by

$$\rho = k\left(1 + \frac{y}{a}\right)$$

where k is a positive constant.

(a) By considering horizontal strips, show that $M = \frac{20}{3}ka^2$. **[5]**

(b) Use integration to find the coordinates of the centre of mass of the lamina, giving your answers in terms of a. **[7]**

(c) The lamina is freely suspended from A and hangs in equilibrium. Find the angle between AB and the vertical. **[3]**

Solution

(a) The line OA passes through O and $A(4a, 2a)$, so its equation is $y = \frac{1}{2}x$, that is $x = 2y$. A horizontal strip at height y, of thickness δy, runs from $x = 2y$ to $x = 4a$, so its length is $4a - 2y$. Its density depends only on y, so its mass is

$$\delta m \approx \rho(4a - 2y)\,\delta y = k\left(1 + \frac{y}{a}\right)(4a - 2y)\,\delta y$$

Summing the strips from $y = 0$ to $y = 2a$ and letting $\delta y \to 0$:

$$\begin{aligned} M &= \int_0^{2a} k\left(1 + \frac{y}{a}\right)(4a - 2y)\,\mathrm{d}y \\ &= k\int_0^{2a}\left(4a + 2y - \frac{2y^2}{a}\right)\mathrm{d}y \\ &= k\left[4ay + y^2 - \frac{2y^3}{3a}\right]_0^{2a} \\ &= k\left(8a^2 + 4a^2 - \frac{16a^2}{3}\right) \\ &= \frac{20}{3}ka^2 \end{aligned}$$

as required.

(b) The density of each strip is constant along its length, so the centre of mass of the strip at height y is at its midpoint, where

$$x = \frac{2y + 4a}{2} = y + 2a$$

Taking moments of the strips about the y-axis:

$$\begin{aligned} M\bar{x} &= \int_0^{2a} (y + 2a)\, k\Big(1 + \frac{y}{a}\Big)(4a - 2y)\, \mathrm{d}y \\ &= k \int_0^{2a} \Big(1 + \frac{y}{a}\Big)\big(8a^2 - 2y^2\big)\mathrm{d}y \\ &= k \int_0^{2a} \left(8a^2 + 8ay - 2y^2 - \frac{2y^3}{a}\right)\mathrm{d}y \\ &= k\left[8a^2 y + 4ay^2 - \frac{2y^3}{3} - \frac{y^4}{2a}\right]_0^{2a} \\ &= k\left(16a^3 + 16a^3 - \frac{16a^3}{3} - 8a^3\right) \\ &= \frac{56}{3}ka^3 \end{aligned}$$

Using part (a):

$$\bar{x} = \frac{\frac{56}{3}ka^3}{\frac{20}{3}ka^2} = \frac{14a}{5}$$

Taking moments of the strips about the x-axis:

$$\begin{aligned} M\bar{y} &= \int_0^{2a} y\, k\Big(1 + \frac{y}{a}\Big)(4a - 2y)\, \mathrm{d}y \\ &= k \int_0^{2a} \left(4ay + 2y^2 - \frac{2y^3}{a}\right)\mathrm{d}y \\ &= k\left[2ay^2 + \frac{2y^3}{3} - \frac{y^4}{2a}\right]_0^{2a} \\ &= k\left(8a^3 + \frac{16a^3}{3} - 8a^3\right) \\ &= \frac{16}{3}ka^3 \end{aligned}$$

so

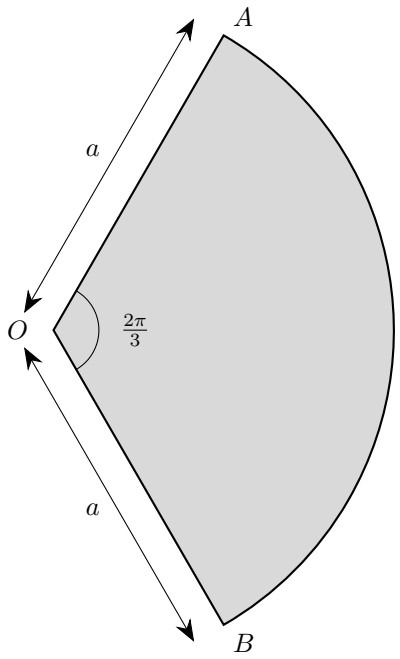
$$\bar{y} = \frac{\frac{16}{3}ka^3}{\frac{20}{3}ka^2} = \frac{4a}{5}$$

The centre of mass is at $\left(\frac{14a}{5}, \frac{4a}{5}\right)$.

(c) When the lamina hangs freely from A, the centre of mass G lies vertically below A, so the required angle θ is the angle between AB and AG. With $A(4a, 2a)$ and $G\left(\frac{14a}{5}, \frac{4a}{5}\right)$, the perpendicular distance of G from AB is $4a - \frac{14a}{5} = \frac{6a}{5}$, and G is $2a - \frac{4a}{5} = \frac{6a}{5}$ below A measured parallel to AB. Hence

$$\tan\theta = \frac{\frac{6a}{5}}{\frac{6a}{5}} = 1 \quad \Longrightarrow \quad \theta = 45^\circ$$

8. *[In this question, you may assume that the centre of mass of a uniform circular arc of radius r, with angle at the centre 2α, is at a distance $\frac{r\sin\alpha}{\alpha}$ from the centre.]*



A thin non-uniform plate is in the shape of a sector OAB of a circle with centre O and radius a. The angle $AOB = \frac{2\pi}{3}$, as shown in the diagram above.
The plate is modelled as a non-uniform lamina. The mass per unit area of the lamina at any point P of the lamina is modelled as $k(2a - OP)$, where $k = \frac{3\lambda}{\pi a^3}$ and λ is a positive constant.
Using the model,

(a) find the mass of the plate in terms of λ, **[5]**

(b) show that the distance of the centre of mass of the plate from O is $\frac{15\sqrt{3}\,a}{16\pi}$. **[4]**

The plate is freely suspended from A and hangs in equilibrium.

(c) Find the angle between OA and the vertical, giving your answer in degrees to 1 decimal place. **[3]**

Solution

(a) Divide the plate into thin circular arcs centred at O. The arc of radius r and thickness δr has length $\frac{2\pi}{3}r$, so its area is approximately $\frac{2\pi}{3}r\,\delta r$. Every point of this arc is at distance r from O, so its mass per unit area is $k(2a - r)$. The mass of the arc is therefore

$$\delta m \approx k(2a - r) \times \frac{2\pi}{3} r\,\delta r$$

Summing over $0 \le r \le a$ and letting $\delta r \to 0$:

$$\begin{aligned} M &= \frac{2\pi k}{3}\int_0^a \big(2ar - r^2\big)\mathrm{d}r \\ &= \frac{2\pi k}{3}\left[ar^2 - \frac{r^3}{3}\right]_0^a \\ &= \frac{2\pi k}{3} \times \frac{2a^3}{3} \\ &= \frac{4\pi k a^3}{9} \end{aligned}$$

Substituting $k = \frac{3\lambda}{\pi a^3}$:

$$M = \frac{4\pi a^3}{9} \times \frac{3\lambda}{\pi a^3} = \frac{4\lambda}{3}$$

(b) The plate is symmetrical about the bisector of angle AOB, so its centre of mass G lies on this line. Let $OG = \bar{x}$.

Each thin arc from part (a) is uniform, with radius r and angle $2\alpha = \frac{2\pi}{3}$ at the centre, so $\alpha = \frac{\pi}{3}$. Its centre of mass lies on the bisector at distance

$$\frac{r\sin\frac{\pi}{3}}{\frac{\pi}{3}} = \frac{3}{\pi} \times \frac{\sqrt{3}}{2}r = \frac{3\sqrt{3}\,r}{2\pi}$$

from O. Taking moments about O along the bisector:

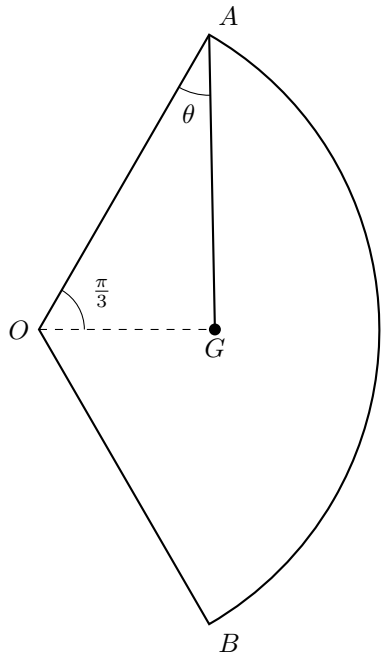
$$\begin{aligned} M\bar{x} &= \int_0^a \frac{3\sqrt{3}\,r}{2\pi} \times k(2a - r)\frac{2\pi}{3}r\,\mathrm{d}r \\ &= \sqrt{3}\,k\int_0^a \big(2ar^2 - r^3\big)\mathrm{d}r \\ &= \sqrt{3}\,k\left[\frac{2ar^3}{3} - \frac{r^4}{4}\right]_0^a \\ &= \sqrt{3}\,k\left(\frac{2}{3} - \frac{1}{4}\right)a^4 \\ &= \frac{5\sqrt{3}\,ka^4}{12} \end{aligned}$$

Using $M = \frac{4\pi k a^3}{9}$ from part (a):

$$\bar{x} = \frac{5\sqrt{3}\,ka^4}{12} \times \frac{9}{4\pi k a^3} = \frac{45\sqrt{3}\,a}{48\pi} = \frac{15\sqrt{3}\,a}{16\pi}$$

as required.

(c) When the plate hangs freely from A, its centre of mass G is vertically below A, so AG is vertical and the required angle is $\theta = \angle OAG$, as in the diagram below.

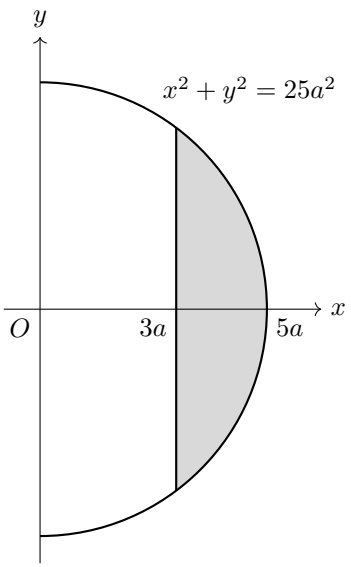


In triangle OAG, $OA = a$, $OG = \bar{x} = \frac{15\sqrt{3}\,a}{16\pi}$ and $\angle AOG = \frac{\pi}{3}$. The components of $\overrightarrow{OG}$ along OA and perpendicular to OA are $\bar{x}\cos\frac{\pi}{3}$ and $\bar{x}\sin\frac{\pi}{3}$, so

$$\tan\theta = \frac{\bar{x}\sin\frac{\pi}{3}}{a - \bar{x}\cos\frac{\pi}{3}} \approx \frac{0.44762a}{0.74156a} \approx 0.60362$$

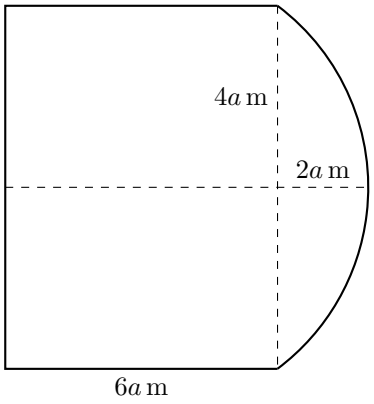
which gives

$$\theta = 31.1^\circ \qquad (1\text{ d.p.})$$



9. The shaded region shown in the diagram above is bounded by the line with equation $x = 3a$ and the curve with equation $x^2 + y^2 = 25a^2$.
This shaded region is rotated through 180° about the x-axis to form a solid of revolution.
This solid is used to model a dome with height $2a$ m and base radius $4a$ m.
The dome is modelled as being non-uniform, with the mass per unit volume of the dome at the point (x, y, z) equal to $\frac{\lambda}{x^2}$ kg m^{-3}, where $3a \leq x \leq 5a$ and λ is a positive constant.

(a) Show that the distance of the centre of mass of the dome from the centre of its plane face is $\left(\frac{75}{4} \ln \frac{5}{3} - 9\right) a$ m. **[6]**



A uniform solid right circular cylinder has radius $4a$ m and height $6a$ m. A toy is formed by joining the plane face of the dome to one plane face of the cylinder, as shown in the diagram above.
The weight of the cylinder is kW and the weight of the dome is $4W$.
The centre of mass of the toy is a distance d m from the plane face of the dome.

(b) Show that

$$d = \frac{\left|3k + 36 - 75 \ln \frac{5}{3}\right|}{k + 4} a$$ **[4]**

The toy is suspended from a point on the circumference of the plane face of the cylinder that is not joined to the dome, and hangs freely in equilibrium with its axis of symmetry at an angle β to the vertical.

(c) Given that $\tan \beta = \frac{4}{5}$, find the exact value of k. **[4]**

Solution

(a) Rotating the shaded region through 180° about the x-axis gives the same solid as rotating the part above the x-axis through 360°. At position x, the cross-section is therefore a circle of radius y, where $y^2 = 25a^2 - x^2$. By symmetry, the centre of mass lies on the x-axis. The density $\rho = \frac{\lambda}{x^2}$ depends only on x. For a solid formed by rotating a region about the x-axis, the mass m_D of the dome and the position $\bar{x}_D$ of its centre of mass satisfy

$$m_D = \pi \int_{3a}^{5a} \rho y^2 \, \mathrm{d}x, \qquad m_D \bar{x}_D = \pi \int_{3a}^{5a} \rho x y^2 \, \mathrm{d}x$$

For the mass:

$$\begin{aligned} m_D &= \pi \int_{3a}^{5a} \frac{\lambda}{x^2}\left(25a^2 - x^2\right)\mathrm{d}x \\ &= \pi\lambda \int_{3a}^{5a} \left(\frac{25a^2}{x^2} - 1\right)\mathrm{d}x \\ &= \pi\lambda \left[-\frac{25a^2}{x} - x\right]_{3a}^{5a} \\ &= \pi\lambda \left((-5a - 5a) - \left(-\frac{25a}{3} - 3a\right)\right) \\ &= \frac{4\pi\lambda a}{3} \,\mathrm{kg} \end{aligned}$$

For the moment, where $x > 0$ throughout, so $\ln x$ needs no modulus:

$$\begin{aligned} m_D \bar{x}_D &= \pi \int_{3a}^{5a} \frac{\lambda}{x^2}\, x\left(25a^2 - x^2\right)\mathrm{d}x \\ &= \pi\lambda \int_{3a}^{5a} \left(\frac{25a^2}{x} - x\right)\mathrm{d}x \\ &= \pi\lambda \left[25a^2 \ln x - \frac{1}{2}x^2\right]_{3a}^{5a} \\ &= \pi\lambda \left(25a^2 \ln(5a) - 25a^2 \ln(3a) - \frac{25a^2}{2} + \frac{9a^2}{2}\right) \\ &= \pi\lambda a^2 \left(25 \ln \frac{5}{3} - 8\right) \end{aligned}$$

Dividing:

$$\begin{aligned} \bar{x}_D &= \frac{\pi\lambda a^2 \left(25 \ln \frac{5}{3} - 8\right)}{\frac{4}{3}\pi\lambda a} \\ &= \frac{3a}{4}\left(25 \ln \frac{5}{3} - 8\right) \\ &= \left(\frac{75}{4} \ln \frac{5}{3} - 6\right) a \end{aligned}$$

The plane face of the dome lies in $x = 3a$, with its centre at $(3a, 0)$, so the distance of the centre of mass from the centre of the plane face is

$$\bar{x}_D - 3a = \left(\frac{75}{4} \ln \frac{5}{3} - 9\right) a \,\mathrm{m}$$

as required.

(b) Keep the same x-axis along the axis of symmetry of the toy. The dome's plane face is at $x = 3a$, so the cylinder occupies $-3a \leq x \leq 3a$. The cylinder is uniform, so its centre of mass is at its centre, $x = 0$. The dome's centre of mass is at $x = \bar{x}_D$ from part (a). For the toy:

$$M\bar{x} = \sum m_i x_i, \quad \text{where } M = \sum m_i$$

Mass is proportional to weight, so the weights kW and $4W$ can be used in place of the masses:

$$(kW + 4W)\bar{x} = kW \times 0 + 4W\left(\frac{75}{4}\ln\frac{5}{3} - 6\right)a$$

$$\bar{x} = \frac{\left(75\ln\frac{5}{3} - 24\right)a}{k+4}$$

The plane face of the dome is at $x = 3a$, so

$$\begin{aligned} d &= |\bar{x} - 3a| \\ &= \frac{\left|75\ln\frac{5}{3} - 24 - 3(k+4)\right|}{k+4}\,a \\ &= \frac{\left|75\ln\frac{5}{3} - 36 - 3k\right|}{k+4}\,a \\ &= \frac{\left|3k + 36 - 75\ln\frac{5}{3}\right|}{k+4}\,a \end{aligned}$$

as required.

(c) The free plane face of the cylinder is at $x = -3a$, with centre C on the axis. Let P be the point of suspension on its circumference and G the centre of mass of the toy, so $PC = 4a$ and PC is perpendicular to the axis. In equilibrium G lies vertically below P, so the angle between the axis GC and the vertical GP is β, and triangle PCG has a right angle at C.

Since C is at $x = -3a$, $CG = \bar{x} + 3a$. Then

$$\begin{aligned} \tan\beta = \frac{PC}{CG} = \frac{4a}{\bar{x} + 3a} &= \frac{4}{5} \\ \bar{x} + 3a &= 5a \\ \bar{x} &= 2a \end{aligned}$$

Using the expression for $\bar{x}$ from part (b):

$$\begin{aligned} \frac{75\ln\frac{5}{3} - 24}{k+4} &= 2 \\ 75\ln\frac{5}{3} - 24 &= 2k + 8 \\ k &= \frac{75}{2}\ln\frac{5}{3} - 16 \approx 3.16 \end{aligned}$$

10. A non-uniform rod AB has length l and weight W. The weight per unit length of the rod at a distance x from A is proportional to $l + kx$, where k is a positive constant.

(a) Show that the centre of mass of the rod is at a distance $\frac{(3+2k)\,l}{3(2+k)}$ from A. **[6]**

In the case $k = 3$, the rod rests in a vertical plane with the end A on rough horizontal ground and with the rod in contact with a smooth horizontal rail at the point C of the rod, where $AC = \frac{4}{5}l$. The rail is perpendicular to the vertical plane containing the rod, and the rod makes an angle θ with the ground, where $0 < \theta < \frac{\pi}{2}$. The coefficient of friction between the rod and the ground is μ.

(b) Show that, for the rod to be in equilibrium,

$$\mu \geq \frac{3\tan\theta}{1 + 4\tan^2\theta}$$ **[7]**

(c) Hence find the least value of μ for which the rod can rest in equilibrium in this way whatever the value of θ. **[2]**

Solution

(a) Let w be the weight per unit length, so that $w = \lambda(l + kx)$ at a distance x from A, where λ is a positive constant. The weight and centre of mass satisfy

$$W = \int_0^l w\,\mathrm{d}x, \qquad W\bar{x} = \int_0^l xw\,\mathrm{d}x$$

where $\bar{x}$ is the distance of the centre of mass from A. The total weight is

$$\begin{aligned} W &= \int_0^l \lambda(l + kx)\,\mathrm{d}x \\ &= \lambda\left[lx + \tfrac{1}{2}kx^2\right]_0^l \\ &= \lambda\left(l^2 + \tfrac{1}{2}kl^2\right) \\ &= \tfrac{1}{2}\lambda l^2(2 + k) \end{aligned}$$

Taking moments about A:

$$\begin{aligned} W\bar{x} &= \int_0^l x\,\lambda(l + kx)\,\mathrm{d}x \\ &= \lambda\left[\tfrac{1}{2}lx^2 + \tfrac{1}{3}kx^3\right]_0^l \\ &= \lambda\left(\tfrac{1}{2}l^3 + \tfrac{1}{3}kl^3\right) \\ &= \tfrac{1}{6}\lambda l^3(3 + 2k) \end{aligned}$$

Dividing by W:

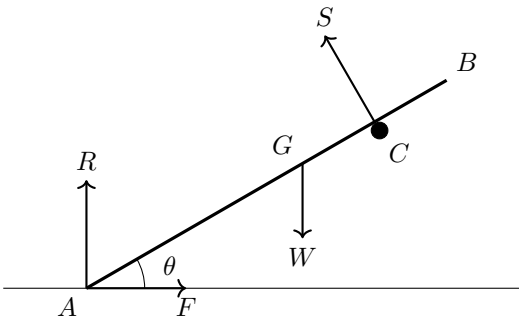
$$\bar{x} = \frac{\frac{1}{6}\lambda l^3(3 + 2k)}{\frac{1}{2}\lambda l^2(2 + k)} = \frac{(3 + 2k)\,l}{3(2 + k)}$$

as required.

(b) With $k = 3$, part (a) gives the centre of mass G at

$$AG = \frac{(3+6)\,l}{3 \times 5} = \frac{3l}{5}$$

The forces on the rod are shown below:



Here R and F are the normal reaction and the friction at A, and S is the reaction of the rail at C, perpendicular to the rod since the rail is smooth. The horizontal component of S acts away from the rail, so the friction at A acts in the opposite direction, as shown.

Taking moments about A:

$$S \times \tfrac{4}{5}l = W \times \tfrac{3}{5}l\cos\theta \quad \Longrightarrow \quad S = \tfrac{3}{4}W\cos\theta$$

Resolving horizontally:

$$F = S\sin\theta = \tfrac{3}{4}W\sin\theta\cos\theta$$

Resolving vertically:

$$\begin{aligned} R + S\cos\theta &= W \\ R &= W - \tfrac{3}{4}W\cos^2\theta \\ &= \tfrac{1}{4}W\left(4 - 3\cos^2\theta\right) \end{aligned}$$

For equilibrium, $F \le \mu R$, so

$$\tfrac{3}{4}W\sin\theta\cos\theta \le \tfrac{1}{4}\mu W\left(4 - 3\cos^2\theta\right)$$

Since $4 - 3\cos^2\theta > 0$, dividing through and then dividing the numerator and denominator by $\cos^2\theta$, using $\sec^2\theta = 1 + \tan^2\theta$:

$$\begin{aligned} \mu &\ge \frac{3\sin\theta\cos\theta}{4 - 3\cos^2\theta} \\ &= \frac{3\tan\theta}{4\sec^2\theta - 3} \\ &= \frac{3\tan\theta}{4 + 4\tan^2\theta - 3} \\ &= \frac{3\tan\theta}{1 + 4\tan^2\theta} \end{aligned}$$

as required.

(c) Let $t = \tan\theta$. As θ ranges over $0 < \theta < \frac{\pi}{2}$, t takes every positive value, so by part (b) the rod can rest

whatever the value of θ provided μ is at least the greatest value of $\frac{3t}{1+4t^2}$ for $t > 0$. Since $(2t-1)^2 \geq 0$, it follows that

$$1 + 4t^2 \geq 4t \quad \Longrightarrow \quad \frac{3t}{1+4t^2} \leq \frac{3t}{4t} = \frac{3}{4}$$

with equality when $t = \frac{1}{2}$. So the greatest value is $\frac{3}{4}$, and the least value of μ is

$$\mu = \tfrac{3}{4}$$